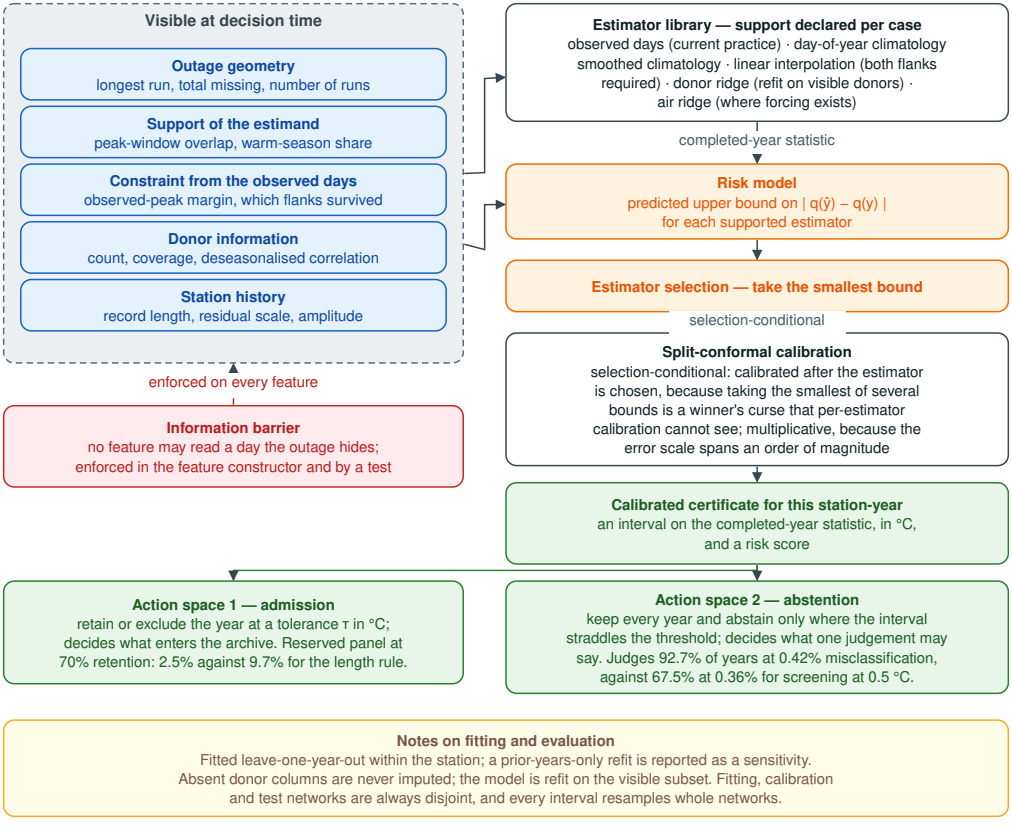
Visible at decision time
Outage geometry
longest run, total missing, number of runs
Support of the estimand
peak-window overlap, warm-season share
Constraint from the observed days
observed-peak margin, which flanks survived
Donor information
count, coverage, deseasonalised correlation
Station history
record length, residual scale, amplitude
enforced on every feature
Information barrier
no feature may read a day the outage hides;
enforced in the feature constructor and by a test
Estimator library — support declared per case
observed days (current practice) · day-of-year climatology
smoothed climatology · linear interpolation (both flanks
required) · donor ridge (refit on visible donors) ·
air ridge (where forcing exists)
completed-year statistic
Risk model
predicted upper bound on | q(ŷ) − q(y) |
for each supported estimator
Estimator selection — take the smallest bound
selection-conditional
Split-conformal calibration
selection-conditional: calibrated after the estimator
is chosen, because taking the smallest of several
bounds is a winner's curse that per-estimator
calibration cannot see; multiplicative, because the
error scale spans an order of magnitude
Calibrated certificate for this station-year
an interval on the completed-year statistic, in °C,
and a risk score
Action space 1 — admission
retain or exclude the year at a tolerance τ in °C;
decides what enters the archive. Reserved panel at
70% retention: 2.5% against 9.7% for the length rule.
Action space 2 — abstention
keep every year and abstain only where the interval
straddles the threshold; decides what one judgement may
say. Judges 92.7% of years at 0.42% misclassification,
against 67.5% at 0.36% for screening at 0.5 °C.
Notes on fitting and evaluation
Fitted leave-one-year-out within the station; a prior-years-only refit is reported as a sensitivity.
Absent donor columns are never imputed; the model is refit on the visible subset. Fitting, calibration
and test networks are always disjoint, and every interval resamples whole networks.